

Public Comment on HHS Health Sector AI RFI (ASTP/ONC)

Patient-Designated Longitudinal Routing to Enable Safe, Effective Clinical AI

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Docket/Document: HHS-ONC-2026-0001-0001

Submission Type: Written comment (public)

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Executive Summary

The RFI correctly identifies that clinical AI adoption is constrained less by model capability than by whether care settings can reliably supply the longitudinal, high-fidelity context needed for safe inference. This comment proposes a structural solution: **patient-designated longitudinal routing by default**.

Core Proposal: Enable a one-time patient designation of a certified longitudinal endpoint to which full-fidelity USCDI data is automatically delivered as it is created, across care settings. This is not a consumer portal strategy. It is a convergence layer that makes complete longitudinal records reliably available back to care delivery systems (with consent) where AI actually runs.

Why This Matters for AI: Clinical AI fails when deployed into informationally impoverished workflows. Patient-designated routing changes the default so that longitudinal completeness becomes the expected input condition for AI, rather than a special-case integration achievement.

Key Recommendations:

1. **Regulation:** Define and certify patient-designated longitudinal endpoints with strict security, consent, provenance, and audit requirements.
2. **Reimbursement:** Pilot payment models tied to verified delivery rather than capability attestations.
3. **R&D:** Fund evaluations that connect longitudinal completeness improvements to clinical outcomes, workflow impact, and AI safety/bias monitoring.

This is a narrow structural change with outsized impact: it improves care coordination immediately, reduces preventable harm, and creates the data conditions necessary for safe, effective, and equitable clinical AI.

1. Introduction: Why I'm Commenting / Relevant Experience

I am submitting this comment because the RFI correctly identifies a core problem: clinical AI adoption is constrained less by model capability than by whether the *care setting can reliably supply the longitudinal, high-fidelity context needed for safe and clinically useful inference*.

My perspective comes from building and operating clinical data infrastructure in real healthcare delivery environments, including clinical data warehouses, integration pipelines, and point-of-care systems used by clinicians under real time constraints. In those settings, the central failure mode is not "no standards exist," but "the record is incomplete when and where clinical decisions are made." AI embedded into workflows inherits that incompleteness, which directly degrades safety, quality, and equity.

2. Background: The Problem the RFI is Trying to Solve

2.1 Fragmentation, not standards maturity, is the binding constraint

Despite FHIR/USCDI maturity, national exchange efforts, and the shift toward API-based access, longitudinal aggregation remains the exception rather than the norm. The operational reality is that records are frequently split across hospitals, specialist groups, labs, urgent care, imaging centers, behavioral health providers, and post-acute settings. Clinicians often have an incomplete view of diagnoses, medications, prior workups, allergies, procedures, problem lists, and results performed elsewhere.

For clinical AI, this is not a "nice-to-have." Longitudinal completeness is the substrate for: accurate clinical context and risk stratification; reduction of false positives/false negatives that arise from missing history; bias mitigation (because missingness is not random and often correlates with socioeconomic access patterns); and trustworthy post-deployment monitoring (drift, subgroup performance, safety event correlation).

2.2 Why fragmentation persists: routing control is mis-assigned

The underlying structural issue is that routing decisions are typically controlled by institutions and vendors, not by patients. Even when standards and exchange exist, **the default routing behavior** tends to produce episodic exchange, partial aggregation, and brittle "islands" of data that do not converge into a reliable longitudinal record accessible at the point of care.

Stated differently: interoperability capability does not guarantee longitudinal completeness, because *the system's routing defaults* do not reliably cause data to converge.

3. Core Proposal: Patient-Designated Longitudinal Routing by Default

3.1 One structural change

HHS can unlock meaningful clinical AI adoption by enabling **patient-designated routing by default**: a one-time patient designation of a certified longitudinal endpoint to which full-fidelity USCDI data is automatically delivered as it is created, across care settings.

Key properties:

4. **One-time designation** (like choosing a primary pharmacy or selecting a health plan PCP)
5. **Automatic delivery** of standardized clinical payloads (USCDI-aligned) as events occur
6. **Persistent routing** via a record-locator mechanism so the designation "follows" the patient
7. **No patient management burden** (no requirement that patients log in, curate, or actively manage data)
8. **Return-to-care enablement**: treating systems can retrieve a current longitudinal view with patient consent, supporting clinician workflow and embedded AI

3.2 What this is not

This is not "a consumer portal strategy" and not a burden-shift to patients. The endpoint is not primarily for human browsing or personal recordkeeping. It is a convergence layer intended to make a complete, current longitudinal record reliably available back to care delivery systems (with consent) where AI actually runs.

3.3 Why this solves the AI adoption bottleneck

Clinical AI fails in practice when deployed into workflows that are informationally impoverished. Patient-designated routing changes the default so that longitudinal completeness becomes the **expected input condition** for AI, rather than a special-case integration achievement.

This is the smallest structural shift that: improves patient safety and care coordination immediately (even without AI); creates the precondition for AI systems to be reliable and monitorable; and aligns incentives in a way enforcement alone has not achieved.

4. Analysis: Impacts, Unintended Consequences, and Why HHS Should Prefer This Approach

4.1 Benefits to clinical care (even before AI)

A persistent longitudinal record reduces: duplicated labs and imaging ordered because prior results are unavailable; medication reconciliation errors (a major source of preventable harm); delays in diagnosis due to missing prior workup and specialist notes; avoidable admissions/readmissions driven by incomplete outpatient history; and clinician time lost hunting for records across portals and faxes. These improvements are direct clinical quality benefits that also improve AI readiness.

4.2 Why incentives matter more than capability attestations

Interoperability has often been framed as "build capability" + "enforce compliance." That approach does not reliably create longitudinal completeness because it does not make completeness an operational default that organizations are paid to produce.

If reimbursement is conditioned on **verified delivery** to a patient-designated endpoint (not mere "capability to deliver"), interoperability shifts from an enforcement problem to an operational norm. This is a durable behavioral change lever.

4.3 Distributional impacts (equity)

Fragmentation and missingness are not evenly distributed. People who receive care across multiple systems (often due to access patterns, insurance churn, transportation constraints, or regional provider availability) are disproportionately harmed by non-convergent records. These same populations are at higher risk of AI underperformance when missingness becomes a hidden input variable correlated with disadvantage.

Patient-designated routing improves equity by making longitudinal completeness a default expectation across sites rather than a privilege of being "in the right health system."

4.4 Semantic interoperability is a real risk—and a necessary forcing function

Implementing default longitudinal routing will reveal known weaknesses in semantic consistency: coding variation, normalization drift, incomplete reconciliation of problem lists, and differing local interpretations. These issues exist today; episodic exchange often masks them.

This should be treated as a forcing function rather than a reason to avoid reform. AI systems amplify semantic inconsistency risks; therefore, operationalizing longitudinal aggregation is the mechanism that makes semantic quality measurable, improvable, and tied to downstream usability.

4.5 Privacy, security, and trust: the "central objection" and how to mitigate it

A convergence endpoint increases the perceived "blast radius" of a breach if designed poorly. That concern is legitimate and should be addressed explicitly in the program design.

Mitigations that can be required through certification and policy guardrails:

- **Consent + revocation:** simple, durable patient consent workflows; immediate revocation capability
- **Strong provenance + auditability:** comprehensive audit logs; patient-accessible "who accessed what and when"
- **Minimum necessary retrieval:** retrieval into treating systems should be scoped; discourage uncontrolled bulk retrieval except when clinically justified
- **Segmentation & sensitive domains:** phased inclusion for highly sensitive categories; require clear labeling and provenance for sensitive information
- **Security baseline:** modern authentication/authorization, encryption in transit/at rest, continuous monitoring, and incident response controls commensurate with handling longitudinal clinical data

A default routing policy must be paired with a certification program that makes "safe aggregation" non-negotiable.

5. Recommended HHS Actions Using the RFI Levers

(Explicitly aligned to the RFI's emphasis on regulation, reimbursement, and R&D.)

5.1 Regulation: Define and certify "longitudinal endpoints" and routing behavior

HHS/ASTP/ONC should define certification criteria for longitudinal endpoints and for EHR behaviors required to support patient designation and delivery.

Certification should require longitudinal endpoints to:

9. Accept complete USCDI-aligned payloads (full-fidelity, not lossy summaries)
10. Support standardized query and return to treating systems with patient consent
11. Provide robust provenance, versioning, and audit logs
12. Enforce strong security controls and clear operational responsibilities
13. Implement consent and revocation mechanisms appropriate for clinical workflows
14. Support resilience and availability expectations appropriate for point-of-care retrieval

Certification should require sending systems to:

15. Support a patient-designated routing instruction
16. Automatically deliver standardized payloads to that endpoint (subject to consent and policy)
17. Support reliable event-based update flows (not just occasional exports)
18. Provide delivery verification (receipt + integrity checks)

This does not require redefining USCDI or replacing FHIR; it changes the default routing behavior and makes longitudinal convergence reliable.

5.2 Persistent routing: record locator mechanism

HHS should establish or enable a persistent routing pointer (record-locator-like) so a patient's designation is discoverable across care settings. That pointer should be patient-controlled, privacy-preserving, and designed to avoid enabling unauthorized tracking.

Conceptually: the routing pointer says "where the longitudinal record lives," not "what is in it." Access to the content remains consent-governed and auditable.

5.3 Reimbursement: tie payment to verified delivery and usability (critical)

Regulation alone will not produce durable change. HHS should pilot reimbursement models where the measure is **successful delivery** (and, over time, usability) rather than capability attestation.

Potential models:

- Infrastructure payments for certified longitudinal endpoints with pricing guardrails

- Quality / value-based measures tied to verified delivery performance (completeness + latency)
- Incentives for treating systems that demonstrate retrieval use in workflows (without penalizing patients)

CMS can directly influence Medicare/Medicaid. Commercial alignment can be approached via demonstrations that show ROI and reduced waste through fewer duplicates and improved coordination.

5.4 R&D and pilots: fund implementation science, not just model research

HHS should sponsor pilots that evaluate:

- **Completeness:** USCDI coverage over time vs expected sources
- **Latency:** time from data creation to endpoint availability
- **Clinical outcomes proxies:** duplicate test reduction, medication discrepancy reduction, preventable adverse events
- **Operational burden:** net clinician time saved vs added workflows
- **AI safety and bias:** subgroup performance changes when longitudinal completeness improves
- **Total cost of ownership:** endpoint ops, integration costs, and security compliance

Study design suggestion: stepped-wedge rollout across sites or regions to enable causal inference without requiring all participants to switch simultaneously.

6. Addressing Likely Counterarguments

Counterargument A: "This is too hard / too expensive"

The current model externalizes costs into duplicated testing, preventable errors, and administrative burden (fax, portal chasing, manual reconciliation). Patient-designated routing consolidates infrastructure investment into a measurable, reusable utility. By tying reimbursement to verified delivery and using certification to reduce bespoke integrations, the system becomes cheaper over time than the current patchwork of one-off interfaces.

Counterargument B: "Patients won't participate"

Patients already make durable healthcare designations (insurance plans, pharmacies, PCPs, consent forms). The program can be designed so designation occurs at registration, enrollment, or during standard administrative encounters, with minimal friction. Critically: patients do not need to "use an app" for this to work.

Counterargument C: "Privacy risk is unacceptable"

The risk is unacceptable only if the endpoint lacks rigorous controls. This is precisely why certification, auditability, consent, and security baselines must be central requirements. A poorly secured fragmented environment is not inherently safer than a well-governed convergence model; it is simply less visible and less accountable.

7. Specific Requests to HHS (Recap as Actionable Items)

19. **Define and certify** patient-designated longitudinal endpoints with strict security, consent, provenance, and audit requirements
20. **Require support** in certified health IT for patient designation and **automated delivery** of standardized USCDI-aligned data to the designated endpoint
21. **Establish a persistent routing pointer** (record-locator-like) so designations follow patients across care settings
22. **Pilot reimbursement models** tied to verified delivery (and later, usability) rather than capability attestations
23. **Fund evaluations** that connect longitudinal completeness improvements to clinical outcomes, workflow impact, and AI safety/bias monitoring

8. Conclusion

Clinical AI cannot reliably succeed without longitudinal completeness available in real workflows. The most direct way to achieve that completeness is to transfer routing control—by default—from institutions to patients through a one-time designation of a certified longitudinal endpoint, paired with persistent routing and strong consent/audit safeguards.

This is a narrow structural change with outsized impact: it improves care coordination immediately, reduces preventable harm, and creates the data conditions necessary for safe, effective, and equitable clinical AI.

References

HHS Health Sector AI RFI (ASTP/ONC), Docket HHS-ONC-2026-0001:

<https://www.regulations.gov/document/HHS-ONC-2026-0001-0001>

Administrative Procedure Act (public comment process context): 5 U.S.C. § 553